

Unsupervised Learning & Time-Series Analysis & Forecasting

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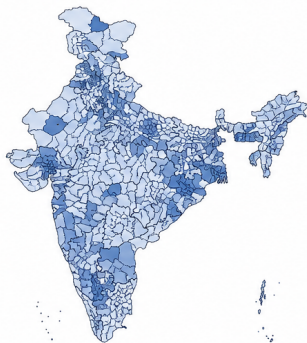
Where Can AI Help in Governance?

Large administrative systems generate enormous amounts of data across many domains. From this data, **two fundamental questions** arise.

Data Generated Across Domains



Across All Districts of India



Massive, Heterogeneous, High-Dimensional Data

Question 1



What patterns are hidden in our data?

Goal: Discover naturally occurring groups or segments without any pre-defined labels.



Unsupervised Learning (Clustering)

Question 2



What is likely to happen next?

Goal: Use historical data to forecast future values and anticipate upcoming trends.



Time-Series Analysis & Forecasting



Outcome: Better insights for planning, prioritization, early warning, resource allocation, and outcome monitoring.

Data-Driven Governance for Better Decisions and Impact.

Unsupervised Learning

Clustering

From raw observations to meaningful groups

Unsupervised Learning

There is no single definition of what a “group” looks like.

K-Means



Idea: partition data around centroids

Best for compact, roughly spherical groups

DBSCAN



Idea: find density-connected groups

Useful for irregular shapes and possible outliers

Hierarchical Clustering



Idea: build nested groups

Useful when hierarchy or multiple levels matter

Gaussian Mixture Models



Idea: model groups probabilistically

Useful when groups overlap and membership is uncertain

From District Data to a Feature Vector

A district becomes one point in a multi-dimensional feature space.

NFHS-5

Gopalganj
Bihar



7 indicators

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Indicator	NFHS-5
Female literacy	63.3%
Women 10+ years schooling	32.9%
Electricity	96.2%
Improved sanitation	60.1%
Clean cooking fuel	39.4%
Institutional births	85.9%
Full vaccination	75.8%

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district profiles

$$\mathbf{x}_d = [x_{d1}, \dots, x_{dp}]$$

one district

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in p dimensions

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Question: Which districts are similar in this feature space?

K-Means: Partition the Data Around Centroids

K-Means procedure

1. Choose K clusters

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Objective

$$J = \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2$$

Minimise within-cluster squared distance.

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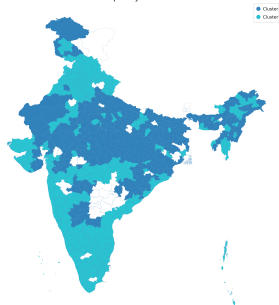
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NFHS-5 result

NFHS-5 Districts Grouped by Similar Profiles — K-Means



707 district profiles $\rightarrow$ 2 clusters

$$\boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$$

The centroid is the mean profile of its cluster.

DBSCAN: Discover Density-Connected Groups

ϵ

Neighbourhood
radius

MinPts

Minimum nearby
points for a dense
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How DBSCAN works

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Repeat for all points.

Core: dense | **Border:** reachable | **Noise:** not reachable

DBSCAN: Discover Density-Connected Groups

ϵ Neighbourhood radius

MinPts Minimum nearby points for a dense neighbourhood

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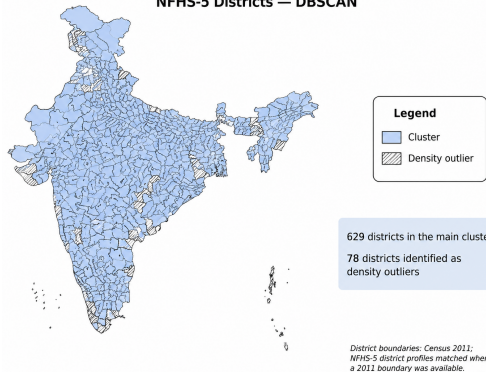
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NFHS-5 result

NFHS-5 Districts — DBSCAN



District profiles grouped according to density

No need to specify the number of clusters in advance.

Time-Series Analysis & Forecasting

From finding structure in data to predicting what comes next

Can We Predict India's Electricity Demand?

A power system must plan for demand before it happens.

Why does this matter?

Capacity planning



Grid operations



Resource allocation

Electricity demand changes over time — and the system must anticipate those changes.

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The question is simple:
What happens next?

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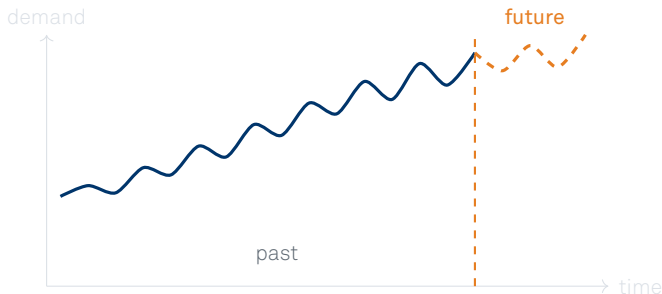


Illustration of the forecasting problem

Can historical demand help us anticipate future demand?

What Is a Time Series?

A time series is a sequence of observations indexed by time.

Time matters

$$y_1, y_2, y_3, \dots, y_t$$

The observations are not interchangeable.

Order carries information.



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→ time

What can be hiding inside the series?

Level

Trend

Seasonality

Noise

typical
magnitude

long-term
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repeating
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unpredictable
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Observed series = structure + noise

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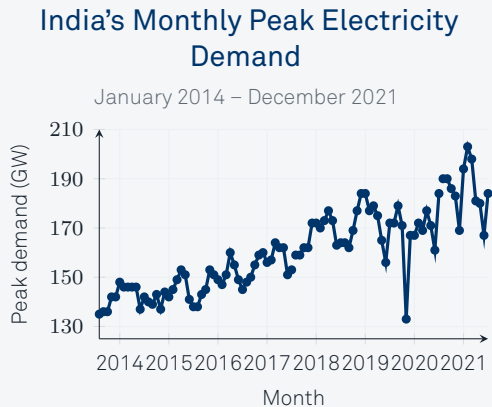
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Forecasting begins by understanding which of these patterns are predictable.

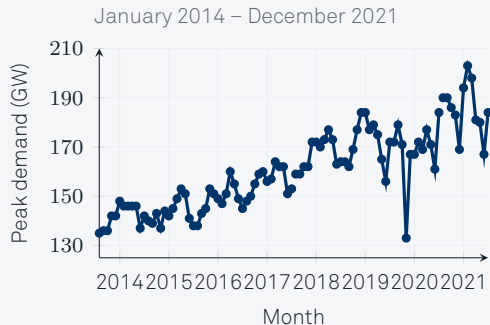
Autocorrelation: The Data Has Memory



Source: CEA, reproduced in BEE's India Energy Scenario 2023–24; values rounded to GW.

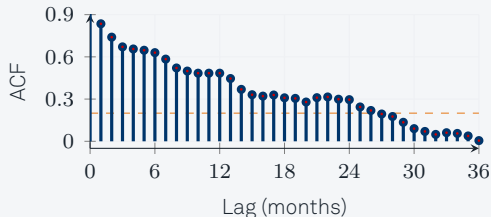
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India's Monthly Peak Electricity Demand



Source: CEA, reproduced in BEE's India Energy Scenario 2023–24; values rounded to GW.

Autocorrelation Function (ACF)



What does this tell us?

- **Lag 1:** strong dependence on the previous month.
- **Lag 12:** the same month last year still carries information.
- **Gradual decay:** dependence persists across many months.

ACF computed from the 96 monthly observations shown at left.

ARIMA: Temporal Dependence to Forecasting

I: Make the series stable

If the level changes over time, model the **change** rather than the level:

$$\Delta y_t = y_t - y_{t-1}$$

Real example from the demand series:



I = Integrated; d = number of times the series is differenced.

AR: Use past values

Past values can help predict the current value of the differenced series.

$$\Delta y_t = c + \phi \Delta y_{t-1} + \varepsilon_t$$

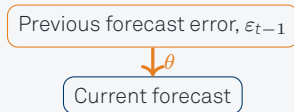


*AR = AutoRegressive;
p = number of AR terms.*

MA: Learn from past errors

Recent forecast errors can help correct the next prediction.

$$\Delta y_t = c + \phi \Delta y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$$



MA = Moving Average; q = number of MA terms.

$$\text{ARIMA}(p, d, q) = \text{AR} + \text{I} + \text{MA}$$

past values | differencing | past errors

ARIMAX: When External Drivers Matter

ARIMA looks inward

ARIMA uses the history of the target series to predict its future.



But electricity demand can also respond to conditions **outside the series itself**.

ARIMA: historical demand $\longrightarrow$ future demand

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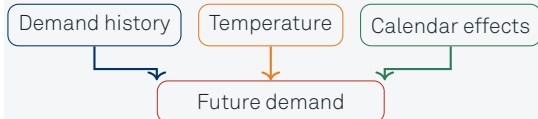


But electricity demand can also respond to conditions **outside the series itself**.

ARIMA: historical demand $\longrightarrow$ future demand

ARIMAX looks outward

Add external variables that may help explain changes in demand.



$$\Delta y_t = c + \phi \Delta y_{t-1} + \beta x_t + \varepsilon_t + \theta \varepsilon_{t-1}$$

x_t = external variable, β = estimated effect of that variable

Hotter weather $\rightarrow$ greater cooling demand $\rightarrow$ higher electricity demand.

ARIMAX = ARIMA + external information

External variables must be known or forecast for future prediction.

SARIMA: Modeling Seasonality

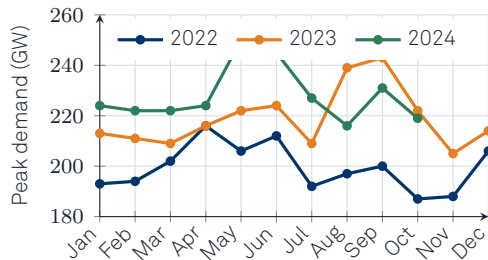
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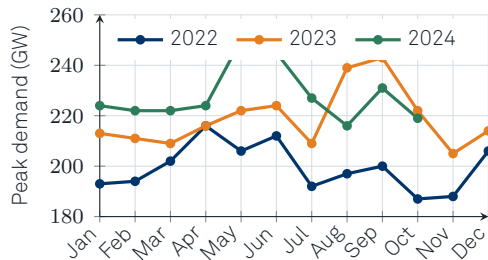


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SARIMA(p,d,q)(P,D,Q)_s
models both ordinary and seasonal dependence.

Recent demand history

Same month last year

Future demand

Seasonal differencing

$$\Delta_{12}y_t = y_t - y_{t-12}$$

Here, the seasonal period is $s = 12$.
Example: January 2019: 164 GW
January 2018: 159 GW

$$\Delta_{12}y_{\text{Jan 2019}} = 164 - 159 = +5 \text{ GW}$$

“How much higher is this January than last January?”

XGBoost: Turning Time into Features

Forecasting becomes supervised learning.

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Build a feature vector

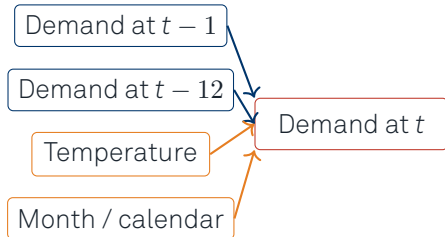
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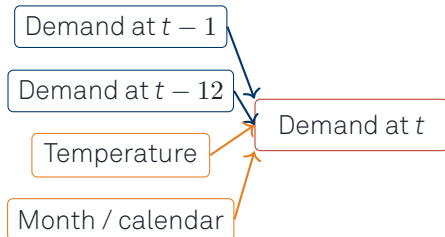


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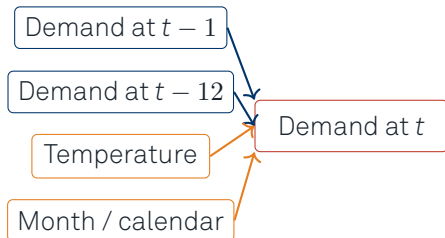
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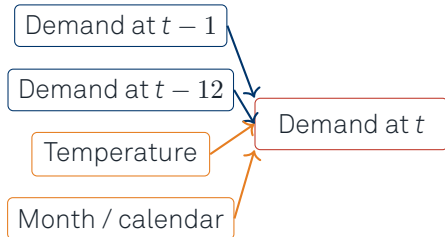
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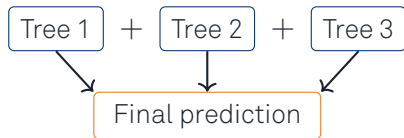
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Each new tree improves the errors left by the previous trees.

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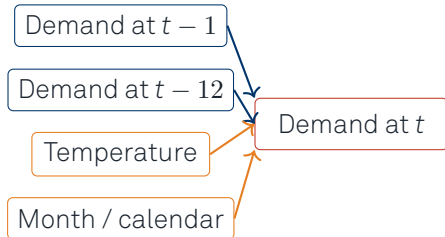
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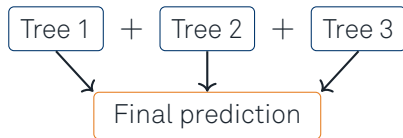
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Why useful for forecasting?

Captures **nonlinear interactions** among lag, seasonal and external features.

Final Takeaway: From Data to Decisions

1. Find the structure

Unsupervised learning

What does a “group” look like?

K-Means | DBSCAN |
Hierarchical

2. Predicting Future

Time-series analysis

What does the past tell us?

ARIMA | SARIMA |
ARIMAX

3. Learn what matters

Supervised learning

Can useful information
become features?

XGBoost
lags + seasonality + drivers

Understand the structure → represent the information → choose the model

Good modeling is not about using the most complex algorithm.

It is about matching the model to the information in the data.

Thank You

Questions & Discussion
